

Module:3 Information Channels

Communication Channels:

A practical Communication system is divided into a transmitter, physical channel or transmission medium and a receiver.

A transmitter consists of an encoder and a modulator, while the receiver consists of a demodulator and a decoder. A channel can be defined as a medium through which the coded signals generated by an information source are transmitted. The Block diagram shows the characterization of a binary communication channel.

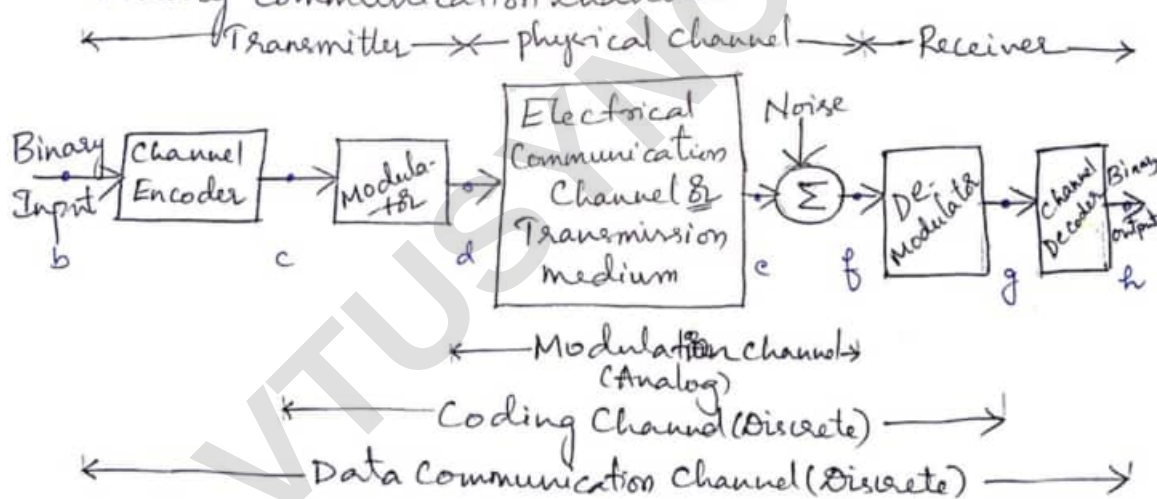


fig. 1. Characterization of a binary Communication Channel.

A points b/w c & g in the system is a discrete channel, often referred to as coding channel, that accepts a sequence of symbols at its input and

produces a sequence of symbols at its output.

The communication channel between points 'd' and 'f' in the system provides the electrical connection between the transmitter & the receiver. The input and output are analog electrical waveforms. This portion of the channel is often called a continuous or modulation channel.

eg: High frequency Radio Systems.

The points between 'b' & 'h', we have binary data being presented at the input & binary data back at the output & hence the channel is called Data Communication Channel (Discrete).

In channel, there are various kinds of disturbances. These disturbances may be due to amplitude and frequency response variations of the channel, and also due to non-linearities in the channel. All these disturbances introduce errors in data transmission and limit the maximum rate at which data can be transferred over the channel.

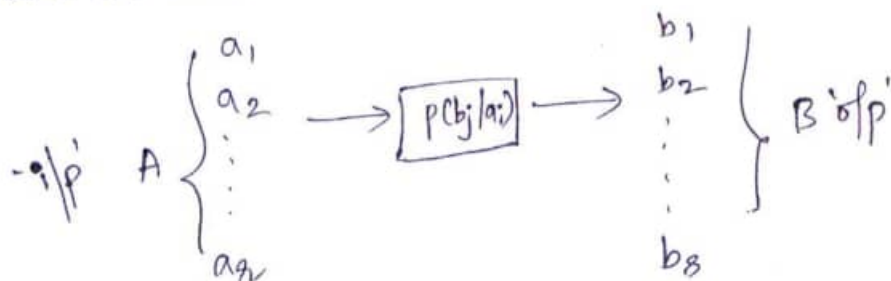
An important characteristic of a data communication system is the "channel capacity" which represents the maximum rate at which data is transferred across the channel with a small probability of error.

Through Shannon-Hartley law, concept of channel capacity of discrete communication channel is studied.

Channel Models:Representation of a channel:

The communication channel between the points 'C' & 'g' is discrete in nature. In general, the input to the channel is a symbol belonging to an alphabet "A" with "x" symbols, the output of the channel is a symbol belonging to some other alphabet "B" with "y" symbols. Due to errors in the channel, the output symbol may be different from the input symbol during any symbol interval. Errors are mainly due to noise in the analog portion of the channel. $\therefore$ The discrete communication channel is modelled in this section.

A communication channel may be represented by $A = \{a_1, a_2, \dots, a_x\}$ i/p with 'x' symbols & $B = \{b_1, b_2, \dots, b_y\}$ o/p with 'y' symbols. & a set of conditional probabilities $P(b_j/a_i)$ with $i = 1, 2, 3, \dots, x$ and $j = 1, 2, \dots, y$ can be shown as below

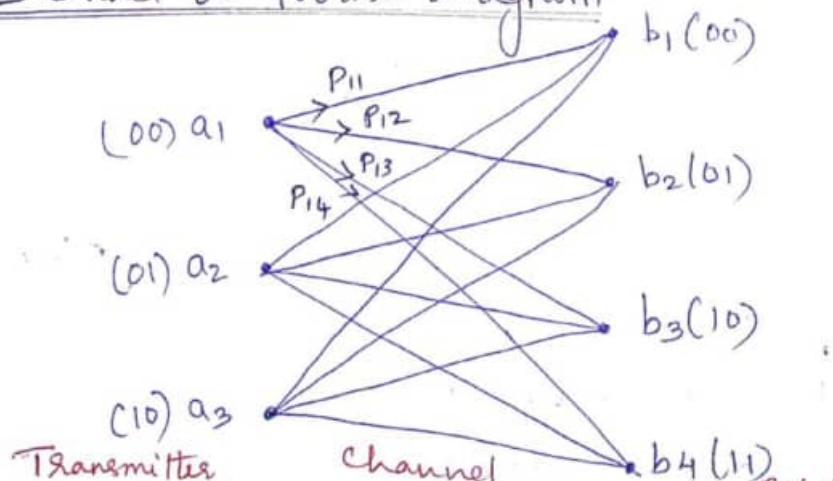


The conditional probabilities came into existence due to the presence of noise in the channel. Because of the noise, there will be some amount of uncertainty about the reception of any symbol. $\therefore$ we have different number of symbols at the receiver & transmi-
-ter is, is & 'x'

$\therefore$ we have 'x x s' no. of conditional probabilities which are represented in a matrix form with i/p symbol row-wise & output symbol column wise & such a matrix is called Channel matrix or Noise matrix is given below.

$$P(b_j|a_i) = P(B/A) = \begin{matrix} & b_1 & b_2 & b_3 & \dots & b_s \\ \begin{matrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_n \end{matrix} & \begin{bmatrix} p(b_1|a_1) & p(b_2|a_1) & p(b_3|a_1) & \dots & p(b_s|a_1) \\ p(b_1|a_2) & p(b_2|a_2) & p(b_3|a_2) & \dots & p(b_s|a_2) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ p(b_1|a_n) & p(b_2|a_n) & \dots & \dots & p(b_s|a_n) \end{bmatrix} \end{matrix}$$

Channel or Noise diagram



Let us consider the channel diagram to understand the effect of noise & intersymbol conversion. Consider a transmitter with $A = \{a_1, a_2, a_3\}$ which are encoded using binary coding as 00, 01 & 10 respectively. Due to noise present in the channel, we may receive four symbols at the receiver: $B = \{b_1, b_2, b_3, b_4\}$ with code words 00, 01, 10 & 11.

Note: We are not transmitting any symbol corresponding to code word "11" at the transmitter but it is received due to noise present in the channel.

In the channel diagram, let us consider a particular case of i/p symbol " a_1 " being transmitted & received as any symbol at the receiver. This "chance" is the conditional probability that the symbol " b_1 " received given that " a_1 " is transmitted at the transmitter denoted by $P(b_1/a_1)$ i.e., $P_{11} = P(b_1/a_1)$, a line connecting a_1 & b_1 in channel diagram.

$$\text{Similarly } P_{12} = P(b_2/a_1), P_{13} = P(b_3/a_1), P_{14} = P(b_4/a_1).$$

$$P_{11} + P_{12} + P_{13} + P_{14} = 1.$$

$$P(b_1/a_1) + P(b_2/a_1) + P(b_3/a_1) + P(b_4/a_1) = 1.$$

Generalizing the above eqⁿ for " s " output symbols & " r " input symbols, we get.

$$P(b_1/a_1) + P(b_2/a_1) + P(b_3/a_1) + \dots + P(b_s/a_1) = 1$$

Writing with summation

$$\sum_{j=1}^s P(b_j/a_1) = 1$$

Above eqⁿ hold good for all "i" input sym

$$\sum_{j=1}^s P(b_j/a_i) = 1$$

From above eqⁿ, it is clear that "Sum of all the elements in any row of the channel matrix is equal to unity."

Let us consider the probabilities of the i/p symbols namely $P(a_1), P(a_2) \dots P(a_s)$ are also known, then

$$P(a_1) + P(a_2) + \dots + P(a_s) = 1$$

$$\therefore \sum_{i=1}^s P(a_i) = 1$$

Knowing the channel matrix elements $P(b_j/a_i)$ for all 'i' & 'j' & the i/p probabilities $P(a_i)$ for all "i", the probabilities of the o/p symbols $P(b_j)$ for all 'j' can be found using theorem of total probability as

$$P(b_j) = P(b_j/a_1)P(a_1) + P(b_j/a_2)P(a_2) + \dots + P(b_j/a_s)P(a_s)$$

In general for 's' output symbols,

$$P(b_s) = P(b_s|a_1) \cdot P(a_1) + P(b_s|a_2) \cdot P(a_2) + \dots - P(b_s|a_n) \cdot P(a_n)$$

Knowing the i/p probabilities $P(a_i)$, Channel matrix elements $P(b_j|a_i)$ & output probabilities $P(b_j)$ input conditional probabilities $P(a_i|b_j)$ can be found using Bayes' Rule.

$$P(a_i|b_j) = \frac{P(b_j|a_i)P(a_i)}{P(b_j)}$$

Joint Probability:

The joint probability between any input symbol " a_i " and any output symbol " b_j " is given by

$$P(a_i, b_j) = P(a_i \cap b_j) = P(b_j|a_i)P(a_i) = P(a_i|b_j) \cdot P(b_j)$$

To obtain the Joint Probability matrix, we multiply all the elements of 1st row of channel matrix by $P(a_1)$, 2nd row by $P(a_2)$ & n^{th} row by $P(a_n)$, we get

$$P(b_j|a_i) \cdot P(a_i) = \begin{matrix} & \begin{matrix} b_1 & b_2 & \dots & b_n \end{matrix} \\ \begin{matrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{matrix} & \begin{bmatrix} P(b_1|a_1)P(a_1) & P(b_2|a_1)P(a_1) & \dots & P(b_n|a_1)P(a_1) \\ P(b_1|a_2)P(a_2) & P(b_2|a_2)P(a_2) & \dots & P(b_n|a_2)P(a_2) \\ \vdots & \vdots & \ddots & \vdots \\ P(b_1|a_n)P(a_n) & P(b_2|a_n)P(a_n) & \dots & P(b_n|a_n)P(a_n) \end{bmatrix} \end{matrix}$$

Replacing by $P(a_i, b_j)$ for $P(b_j|a_i) \cdot P(a_i)$

$$P(a_i, b_j) \text{ or } P(A, B) = \begin{matrix} & \begin{matrix} b_1 & b_2 & \dots & b_s \end{matrix} \\ \begin{matrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{matrix} & \begin{bmatrix} P(a_1, b_1) & P(a_1, b_2) & \dots & P(a_1, b_s) \\ P(a_2, b_1) & P(a_2, b_2) & \dots & P(a_2, b_s) \\ \vdots & \vdots & \ddots & \vdots \\ P(a_n, b_1) & P(a_n, b_2) & \dots & P(a_n, b_s) \end{bmatrix} \end{matrix}$$

The matrix with various joint probabilities b/w input & output symbols is called Joint Probability matrix popularly written as JPM $P(a_i, b_j)$ or $P(A, B)$.

Defⁿ: Joint probability can be defined as the product of output conditional probability & i/p probability or i/p conditional probability & o/p probability.

Properties of JPM:

① Property 1: By adding the elements of JPM columnwise, we can obtain the probability of o/p symbols.

W.K.T

$$P(b_1) = P(b_1|a_1) \cdot P(a_1) + P(b_1|a_2) \cdot P(a_2) + \dots + P(b_1|a_n) \cdot P(a_n)$$

We can rewrite as

$$P(b_1) = P(a_1, b_1) + P(a_2, b_1) + \dots + P(a_n, b_1)$$

i.e., By adding all the elements of 1st column of JPM we get first output symbol 'b₁'.

only for 'x' symbols.

∴ This property can be expressed as.

$$\sum_{i=1}^n P(a_i, b_j) = P(b_j)$$

② Property 2: By adding the elements of JPM rowwise we can obtain the probability of input symbols.

i.e.,

$$P(a_1) = P(a_1|b_1)P(b_1) + \dots + P(a_1|b_s)P(b_s)$$

$$P(a_1) = P(a_1, b_1) + P(a_1, b_2) + \dots + P(a_1, b_s)$$

for 's' symbols

$$P(a_s) = P(a_s, b_1) + P(a_s, b_2) + \dots + P(a_s, b_s)$$

∴ It is observed that, the above all the elements of successive rows of JPM, which are i/p probabilities, can be expressed as

$$\sum_{j=1}^s P(a_i, b_j) = P(a_i)$$

③ Property 3: The sum of all the elements of JPM is equal to unity.

$$\text{i.e., } \sum_{i=1}^n \sum_{j=1}^s P(a_i, b_j) = 1$$

i.e., adding all the elements of property 2, we get the 3rd property of JPM.

P1. In a communication system, a transmitter has 3 input symbols $A = \{a_1, a_2, a_3\}$ and receiver has 3 output symbols $B = \{b_1, b_2, b_3\}$. The matrix given below shows JPM with some marginal probabilities.

$a_i \backslash b_j$	b_1	b_2	b_3
a_1	$\frac{1}{12}$	*	$\frac{5}{36}$
a_2	$\frac{5}{36}$	$\frac{1}{9}$	$\frac{5}{36}$
a_3	*	$\frac{1}{6}$	*

$$P(b_j) \quad \left[\frac{1}{3} \quad \frac{14}{36} \quad * \right]$$

(i) Find the missing probabilities (*) in the table

(ii) Find $P(b_3|a_1)$ & $P(a_1|b_3)$

(iii) Are the events a_1 & b_1 statistically independent? why?

Sol: (i) W.K.T $\sum_{j=1}^3 P(b_j) = 1$

$$3 = \frac{1}{3} + \frac{14}{36} + P(b_3) = 1$$

$$P(b_3) = \frac{5}{18}$$

From the property of JPM,

$$P(b_1) = P(a_1, b_1) + P(a_2, b_1) + P(a_3, b_1)$$

$$\frac{1}{3} = \frac{1}{12} + \frac{5}{36} + P(a_3, b_1)$$

$$P(a_3, b_1) = \frac{1}{9}$$

$$P(b_2) = P(a_1, b_2) + P(a_2, b_2) + P(a_3, b_2)$$

$$\frac{14}{36} = P(a_1, b_2) + \frac{1}{9} + \frac{1}{6}$$

$$P(a_1, b_2) = \frac{1}{9}$$

$$P(b_3) = P(a_1, b_3) + P(a_2, b_3) + P(a_3, b_3)$$

$$\frac{5}{18} = \frac{5}{36} + \frac{5}{36} + P(a_3, b_3)$$

$$\therefore \boxed{P(a_3, b_3) = 0}$$

(ii) To find $P(b_3|a_1)$ conditional probability we will consider JP ; $P(a_1, b_3) = P(b_3|a_1) \cdot P(a_1)$

w.o.f.f. $P(a_1) = P(a_1, b_1) + P(a_1, b_2) + P(a_1, b_3) \rightarrow 1^{\text{st}} \text{ row.}$

$$P(a_1) = \frac{1}{12} + \frac{1}{9} + \frac{5}{36}$$

$$\boxed{P(a_1) = \frac{1}{3}}$$

$$\therefore P(b_3|a_1) = \frac{P(a_1, b_3)}{P(a_1)} = \frac{5/36}{1/3}$$

$$\therefore \boxed{P(b_3|a_1) = \frac{5}{12}}$$

$$P(a_1|b_3) = \frac{P(a_1, b_3)}{P(b_3)} = \frac{5/36}{5/18}$$

$$\boxed{P(a_1|b_3) = \frac{1}{2}}$$

(iii) For a_1 & b_1 to be statistically independent, consider $P(a_1 \cap b_1) = P(a_1, b_1) = P(a_1)P(b_1)$

$$P(a_1)P(b_1) = \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$$

$$\& P(a_1, b_1) = \frac{1}{12}$$

$$\therefore P(a_1, b_1) \neq P(a_1) \cdot P(b_1)$$

$\therefore a_1$ & b_1 are not statistically independent. b_1 is dependent on a_1 , $\because b_1$ is received by transmitting a_1 .

Entropy Functions and Equivocation

1. Priori Entropy:

The entropy of the input symbols $a_1, a_2, a_3, \dots, a_n$ before their transmission is defined as "priori entropy" denoted by $H(A)$ & is given by

$$H(A) = \sum_{i=1}^n P(a_i) \log \frac{1}{P(a_i)} \text{ bits/sym.}$$

2. Posteriori (conditional) Entropy:

The entropy of the input symbols $a_1, a_2, \dots, a_n$, after transmission & reception of a particular output symbol ' b_j ' is defined as posteriori or conditional entropy denoted by $H(A|b_j)$ & is given by

$$H(A|b_j) = \sum_{i=1}^n P(a_i|b_j) \log \frac{1}{P(a_i|b_j)} \text{ bits/sym.}$$

Since j varies from 1 to s , we have ' s ' no. of conditional entropies.

3. Equivocation:

It is defined as the average value of all the conditional entropies specified in above equation when ' j ' is varied from 1 to s .

It is denoted by $H(A/B)$ & is given by

$$\begin{aligned} H(A/B) &= E[H(A|b_j)] \\ &= \sum_{j=1}^s P(b_j) \cdot H(A|b_j) \end{aligned}$$

$$\begin{aligned}
 &= \sum_{j=1}^8 P(b_j) \cdot \sum_{i=1}^9 P(a_i|b_j) \log \frac{1}{P(a_i|b_j)} \\
 &= \sum_{i=1}^9 \sum_{j=1}^8 P(b_j) P(a_i|b_j) \log \frac{1}{P(a_i|b_j)} \\
 &= \sum_{i=1}^9 \sum_{j=1}^8 P(a_i, b_j) \log \frac{1}{P(a_i|b_j)} ; \text{ By def of Joint Probability}
 \end{aligned}$$

$$H(A/B) = \sum_{i=1}^9 \sum_{j=1}^8 P(a_i, b_j) \log \frac{1}{P(a_i|b_j)} \text{ bits/sym.}$$

It is a measure of uncertainty, when symbols are transmitted over the channel & hence represent the amount of information lost due to noise with respect to any of the output symbol.

Interchanging A & B ; we get

$$H(B/A) = \sum_{j=1}^8 \sum_{i=1}^9 P(b_j, a_i) \log \frac{1}{P(b_j|a_i)}$$

$$\text{but } P(b_j, a_i) = P(a_i, b_j)$$

$$H(B/A) = \sum_{i=1}^9 \sum_{j=1}^8 P(a_i, b_j) \log \frac{1}{P(b_j|a_i)} \text{ bits/sym.}$$

$H(B/A)$ is also called equivocation, which is a measure of information about the receiver.

Equivocation specifies the average amount of information needed to specify an input symbol provided we are allowed to observe output produced by that input symbol.

Mutual Information:

When an average information of $H(A)$ is transmitted over the channel, an average amount of information equal to equivocation $H(A/B)$ is lost in the channel due to intersymbol conversion which is due to noise. The balance of information received at the receiver with respect to an observed output symbol is called mutual information & is given by

$$H(A) - H(A/B) = I(A, B)$$

Expression for mutual Information in terms of probabilities

$$i.e., I(A, B) = \sum_{i=1}^n P(a_i) \log \frac{1}{P(a_i)} - \sum_{i=1}^n \sum_{j=1}^s P(a_i, b_j) \log \frac{1}{P(a_i | b_j)}$$

$$L.K.T \quad \sum_{j=1}^s P(b_j | a_i) = 1$$

Multiply the 1st term of above eqⁿ by 1 & substitute the above expression & simplify

$$I(A, B) = \sum_{i=1}^n P(a_i) \log \frac{1}{P(a_i)} \left[\sum_{j=1}^s P(b_j | a_i) \right] - \sum_{i=1}^n \sum_{j=1}^s P(a_i, b_j) \log \frac{1}{P(a_i | b_j)}$$

$$= \sum_{i=1}^n \sum_{j=1}^s \underbrace{P(a_i) \cdot P(b_j | a_i)}_{P(a_i, b_j) \text{ i.e., by Joint probability def.}} \log \frac{1}{P(a_i)} - \sum_{i=1}^n \sum_{j=1}^s P(a_i, b_j) \log \frac{1}{P(a_i | b_j)}$$

$$= \sum_{i=1}^n \sum_{j=1}^s P(a_i, b_j) \left[\log \frac{1}{P(a_i)} + \log P(a_i | b_j) \right]$$

$$\text{but, } P(a_i | b_j) = \frac{P(a_i, b_j)}{P(b_j)}$$

$$\therefore I(A, B) = \sum_{i=1}^n \sum_{j=1}^8 p(a_i, b_j) \left[\log \frac{1}{p(a_i)} + \log \frac{p(a_i, b_j)}{p(b_j)} \right]$$

$$I(A, B) = \sum_{i=1}^n \sum_{j=1}^8 p(a_i, b_j) \log \frac{p(a_i, b_j)}{p(a_i) p(b_j)} \text{ bits/sym}$$

Interchanging A & B, the mutual Infⁿ is identical

$$I(A, B) = I(B, A)$$

$$\text{Similarly } I(B, A) = H(B) - H(B/A)$$

$$I(A, B) = H(A) - H(A/B)$$

$H(B)$ is called Entropy of the output symbols given by

$$H(B) = \sum_{j=1}^8 p(b_j) \log \frac{1}{p(b_j)} \text{ bits/sym.}$$

We can also obtain Joint Entropy from mutual information as,

$$I(A, B) = I(B, A)$$

$$H(A) - H(A/B) = H(B) - H(B/A)$$

$$H(A) + H(B/A) = H(B) + H(A/B)$$

$$H(A, B) = H(B, A)$$

i.e., $H(A, B)$ & $H(B, A)$ is defined as Joint Entropy

The joint Entropy in terms of probability can be obtained as,

$$H(A, B) = \sum_{i=1}^n p(a_i) \log \frac{1}{p(a_i)} + \sum_{i=1}^n \sum_{j=1}^8 p(a_i, b_j) \log \frac{1}{p(b_j|a_i)}$$

$$\text{W.K.T. } \sum_{j=1}^8 p(b_j|a_i) = 1, \text{ substituting in 1st part of above eq}$$

$$\begin{aligned}
 &= \sum_{i=1}^n \sum_{j=1}^s \underbrace{p(a_i) p(b_j|a_i)}_{p(a_i, b_j) \text{ by joint probability def.}} \log \frac{1}{p(a_i)} + \sum_{j=1}^s \sum_{i=1}^n \underbrace{p(a_i, b_j)}_{p(b_j|a_i)} \log \frac{1}{p(b_j|a_i)} \\
 &= \sum_{i=1}^n \sum_{j=1}^s p(a_i, b_j) \log \frac{1}{p(a_i)} + \sum_{i=1}^n \sum_{j=1}^s p(a_i, b_j) \log \frac{1}{p(b_j|a_i)} \\
 &= \sum_{i=1}^n \sum_{j=1}^s p(a_i, b_j) \left[\log \frac{1}{p(a_i)} + \log \frac{1}{p(b_j|a_i)} \right] \\
 &= \sum_{i=1}^n \sum_{j=1}^s p(a_i, b_j) \log \frac{1}{p(a_i) \cdot p(b_j|a_i)} \\
 &\quad \underbrace{p(a_i, b_j)}_{\text{by J.P def.}} \\
 \boxed{H(A, B) = \sum_{i=1}^n \sum_{j=1}^s p(a_i, b_j) \log \frac{1}{p(a_i, b_j)} \text{ bits/sym.}}
 \end{aligned}$$

Properties of Mutual Information:

We have obtained interrelationship between various entropies and also mutual information.

Based on these interrelationships, we list few properties

- ① The mutual information of a channel is symmetric
i.e., $I(A, B) = I(B, A)$
- ② The mutual information is always positive.
i.e., $I(A, B) \geq 0$
- ③ The mutual information of a channel may be expressed in terms of entropy of the channel output also as,
 $I(A, B) = H(B) - H(B|A)$.

④ The mutual information is related to the joint entropy of the channel as,

$$I(A, B) = H(A) - H(A|B)$$

$$I(B, A) = H(B) - H(B|A)$$

$$H(B|A) = H(B) - I(B, A)$$

$$H(B|A) = H(B) - I(A, B) \text{ substituting in}$$

$$H(A) + H(B|A) = H(A, B), \text{ we get.}$$

$$H(A) + H(B) - I(A, B) = H(A, B)$$

$$\underline{\underline{\text{Q.E.D.}}} \quad \boxed{I(A, B) = H(A) + H(B) - H(A, B)}$$

P1. Show that $H(X, Y) = H(X|Y) + H(Y)$

$$\text{L.H.S. } H(X|Y) = \sum_{i=1}^n \sum_{j=1}^8 p(x_i, y_j) \log \frac{1}{p(x_i|y_j)}$$

$$\& \quad H(Y) = \sum_{j=1}^8 p(y_j) \log \frac{1}{p(y_j)}$$

Consider,

$$\text{L.H.S. } H(X, Y) = \sum_{i=1}^n \sum_{j=1}^8 p(x_i, y_j) \log \frac{1}{p(x_i, y_j)}$$

$$= \sum_{i=1}^n \sum_{j=1}^8 p(x_i, y_j) \log \frac{1}{p(x_i|y_j) \cdot p(y_j)} \quad \text{By J.P def.}$$

$$= \sum_{i=1}^n \sum_{j=1}^8 p(x_i, y_j) \log \frac{1}{p(x_i|y_j)} + \sum_{i=1}^n \sum_{j=1}^8 p(x_i, y_j) \log \frac{1}{p(y_j)}$$

$$= H(X|Y) + \sum_{j=1}^8 \left[\sum_{i=1}^n p(x_i, y_j) \right] \log \frac{1}{p(y_j)}$$

$$\text{By property of JPM } \downarrow \sum_{i=1}^n p(x_i, y_j) = p(y_j)$$

$$= H(X/Y) + \sum_{j=1}^3 p(y_j) \log \frac{1}{p(y_j)}$$

$$\therefore H(X, Y) = H(X/Y) + H(Y) \quad \text{Hence proved.}$$

P2. For the JPM given below, compute individually $H(X)$, $H(Y)$, $H(X, Y)$, $H(X/Y)$, $H(Y/X)$ & $I(X, Y)$. Verify the relationship among these entropies.

$$P(X, Y) = \begin{bmatrix} 0.05 & 0 & 0.20 & 0.05 \\ 0 & 0.10 & 0.10 & 0 \\ 0 & 0 & 0.20 & 0.10 \\ 0.05 & 0.05 & 0 & 0.10 \end{bmatrix} \quad \begin{array}{l} \text{JPM} \\ \text{joint probability} \\ \text{matrix} \end{array}$$

Sol: From the properties of JPM;
Summation of rows gives input probabilities

$$P(X_1) = 0.05 + 0.2 + 0.05 = 0.3$$

$$P(X_2) = 0.10 + 0.10 = 0.2$$

$$P(X_3) = 0.2 + 0.1 = 0.3$$

$$P(X_4) = 0.05 + 0.05 + 0.1 = 0.2$$

Summation of column elements gives o/p probabilities

$$P(Y_1) = 0.05 + 0.05 = 0.1$$

$$P(Y_2) = 0.1 + 0.05 = 0.15$$

$$P(Y_3) = 0.2 + 0.1 + 0.2 = 0.5$$

$$P(Y_4) = 0.05 + 0.1 + 0.1 = 0.25$$

$$\text{Entropy of input symbols; } H(X) = \sum_{i=1}^4 P(X_i) \log \frac{1}{P(X_i)}$$

$$H(X) = 2(0.3 \log \frac{1}{0.3}) + 2(0.2 \log \frac{1}{0.2}) = \underline{\underline{1.971 \text{ bits/sym}}}$$

Entropy of o/p symbols; $H(Y) = \sum_{j=1}^4 p(y_j) \log \frac{1}{p(y_j)}$

$$H(Y) = 0.1 \log \frac{1}{0.1} + 0.15 \log \frac{1}{0.15} + 0.5 \log \frac{1}{0.5} + 0.25 \log \frac{1}{0.25}$$

$$H(Y) = \underline{1.743 \text{ bits/sym}}$$

Joint entropy: $H(X, Y) = \sum_{i=1}^4 \sum_{j=1}^4 p(x_i, y_j) \log \frac{1}{p(x_i, y_j)}$

$$= \left(0.05 \log \frac{1}{0.05}\right) 4 + \left(0.1 \log \frac{1}{0.1}\right) 4 + \left(0.2 \log \frac{1}{0.2}\right) 2$$

$$H(X, Y) = \underline{3.122 \text{ bits/sym}}$$

Equivocation; $H(X/Y) = \sum_{i=1}^4 \sum_{j=1}^4 p(x_i, y_j) \log \frac{1}{p(x_i/y_j)}$

$$p(X/Y) = \frac{p(X, Y)}{p(Y)} ; \text{channel matrix:}$$

$$\therefore p(X/Y) = \begin{bmatrix} \frac{0.05}{0.1} & 0 & \frac{0.2}{0.5} & \frac{0.05}{0.25} \\ 0 & \frac{0.10}{0.15} & \frac{0.1}{0.5} & 0 \\ 0 & 0 & \frac{0.2}{0.5} & \frac{0.1}{0.25} \\ \frac{0.05}{0.1} & \frac{0.05}{0.15} & 0 & \frac{0.1}{0.25} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 & \frac{2}{5} & \frac{1}{5} \\ 0 & \frac{2}{3} & \frac{1}{5} & 0 \\ 0 & 0 & \frac{2}{5} & \frac{2}{5} \\ \frac{1}{2} & \frac{1}{3} & 0 & \frac{2}{5} \end{bmatrix}$$

$$H(X/Y) = \underline{1.379 \text{ bits/sym}}$$

$$\text{Equivocation } H(Y/X) = \sum_{i=1}^4 \sum_{j=1}^4 p(x_i, y_j) \cdot \log \frac{1}{p(y_j/x_i)}$$

$$P(Y/X) = \frac{P(X,Y)}{P(X)} = \begin{bmatrix} \frac{0.05}{0.3} & 0 & \frac{0.2}{0.3} & \frac{0.05}{0.3} \\ 0 & \frac{0.1}{0.2} & \frac{0.1}{0.2} & 0 \\ 0 & 0 & \frac{0.2}{0.3} & \frac{0.1}{0.3} \\ \frac{0.05}{0.2} & \frac{0.05}{0.2} & 0 & \frac{0.1}{0.2} \end{bmatrix}$$

$$P(Y/X) = \begin{bmatrix} \frac{1}{6} & 0 & \frac{2}{3} & \frac{1}{6} \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{2} \end{bmatrix}$$

$$H(Y/X) = \underline{1.151} \text{ bits/sym}$$

Verification; $H(Y/X) = H(X,Y) - H(X)$
 $= 3.122 - 1.971$

$$H(Y/X) = \underline{1.151} \text{ bits/sym verified}$$

$$H(X/Y) = H(X,Y) - H(Y)$$

$$= 3.122 - 1.743$$

$$= \underline{1.379} \text{ bits/sym verified.}$$

Mutual Information; $I(X,Y) = H(X) - H(X/Y)$

$$I(X,Y) = 1.971 - 1.379 = \underline{0.592} \text{ bits/sym}$$

Verification; $I(X, Y) = H(Y) - H(Y/X)$
 $= 1.743 - 1.151$
 $= \underline{0.592 \text{ bits/sym}} \text{ verified}$

- P3. A transmitter transmits five symbols with probabilities 0.2, 0.3, 0.2, 0.1 & 0.2. Given the channel matrix $P(B/A)$. calculate $H(B)$ & $H(A, B)$

$$P(B/A) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1/4 & 3/4 & 0 & 0 \\ 0 & 1/3 & 2/3 & 0 \\ 0 & 0 & 1/3 & 2/3 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Sol.

$$P(A, B) = \frac{P(B/A)}{P(A)} = \begin{bmatrix} \frac{1}{0.2} & 0 & 0 & 0 \\ \frac{1/4}{0.3} & \frac{3/4}{0.3} & 0 & 0 \\ 0 & \frac{1/3}{0.2} & \frac{2/3}{0.2} & 0 \\ 0 & 0 & \frac{1/3}{0.1} & \frac{2/3}{0.1} \\ 0 & 0 & \frac{1}{0.2} & 0 \end{bmatrix}$$

$$\therefore P(A|B) = \begin{bmatrix} 1/5 & 0 & 0 & 0 \\ 3/40 & 9/40 & 0 & 0 \\ 0 & 1/15 & 2/15 & 0 \\ 0 & 0 & 1/30 & 1/15 \\ 0 & 0 & 1/5 & 0 \end{bmatrix}$$

from the property of JPM; summation of column element gives o/p probability

$$\therefore P(b_1) = \frac{1}{5} + \frac{3}{40} = \frac{11}{40}$$

$$P(b_2) = \frac{9}{40} + \frac{1}{15} = \frac{7}{24}$$

$$P(b_3) = \frac{2}{15} + \frac{1}{30} + \frac{1}{5} = \frac{11}{30}$$

$$P(b_4) = \frac{1}{15}$$

$$H(B) = \sum_{j=1}^4 P(b_j) \log \frac{1}{P(b_j)}$$

$$= \frac{11}{40} \log \frac{40}{11} + \frac{7}{24} \log \frac{24}{7} + \frac{11}{30} \log \frac{30}{11} + \frac{1}{15} \log 15$$

$$H(B) = \underline{\underline{1.822 \text{ bits/sym}}}$$

$$H(A, B) = \sum_{i=1}^4 \sum_{j=1}^4 P(a_i, b_j) \log \frac{1}{P(a_i, b_j)}$$

$$= 2 \left(\frac{1}{5} \log 5 \right) + \frac{3}{40} \log \frac{40}{3} + \frac{9}{40} \log \frac{40}{9} +$$

$$2 \left(\frac{1}{15} \log 15 \right) + \frac{2}{15} \log \frac{15}{2} + \frac{1}{30} \log 30$$

$$H(A, B) = \underline{\underline{2.7653 \text{ bits/sym}}}$$

Rate of Information transmission over a discrete Channel

We have the entropy of input symbols.

$$H(A) = \sum_{i=1}^q P(a_i) \log \frac{1}{P(a_i)} \quad \text{bits/sym}$$

When we consider a discrete memoryless channel accepting symbols at the rate of " q_s " symbols/sec, then the average rate at which information is going into the channel is given by

$$R_{in} = H(A) \cdot q_s \quad \text{bits/sec.}$$

In general, at the receiver, it is not possible to reconstruct the i/p symbol sequence with certainty by operating on receiving sequence. This is due to errors introduced when the s/g's pass through the channel. These errors are in fact, introduced due to the noise present in the channel, thus some amount of infⁿ is lost in the channel due to noise.

This infⁿ which is lost in the channel has been called equivocation $H(A/B)$. Hence the net amount of information which is the mutual information $I(A, B)$ is given by:

$$I(A, B) = H(A) - H(A/B) \quad \text{bits/sym}$$

The average rate of information transmission R_t is

$$R_t = I(A, B) \cdot \eta_s.$$

$$\boxed{R_t = [H(A) - H(A/B)] \eta_s} \quad \text{bits/sec}$$

Note: Since $I(A, B) = I(B, A) = H(B) - H(B/A)$

$$\text{then } R_t = [H(B) - H(B/A)] \cdot \eta_s.$$

Capacity of a discrete memoryless channel.

The capacity of a discrete memoryless noisy channel is defined as the maximum possible rate of information transmission over the channel. The maximum rate of transmission occurs when the source is matched to the channel.

$$\therefore C = \max \{ R_t \}$$

$$C = \max [H(A) - H(A/B)] \cdot \eta_s$$

Shannon's Theorem on Channel Capacity &

Shannon's Second Theorem.

Shannon's theorem on channel capacity states that "when the rate of information transmission $R_t \leq C$, then there exists a coding technique which enables transmission over a channel with a small probability of error, even in the presence of noise in the channel."

$$\text{b.p.T } R_t = [H(A) - H(B)] \text{ } \eta_s \text{ bits/sec.}$$

$$C = \text{Max} \{ [H(A) - H(A/B)] \} \eta_s \text{ bits/sec.}$$

Note ① If $R_t > C$, then reliable transmission of information is not possible without errors. Thus when $R_t > C$, then the errors cannot be controlled by any coding techniques and the probability of error of receiving correct message becomes close to unity.

② Channel efficiency; $\eta_{ch} = \frac{R_t}{C} \times 100 \%$

$$\eta_{ch} = \frac{I(A, B)}{\max[I(A, B)]} \times 100 \%$$

Channel Redundancy; $R_{\eta} = 1 - \eta_{ch}$

Special Channels:

There are several special channels which are of great interest in the field of communication system. Some of them are listed as

- (1) Symmetric & Uniform Channels.
- (2) Binary Symmetric Channels (BSC)
- (3) Binary Erasure Channels (BEC)
- (4) Noisy Channel
- (5) Deterministic Channel
- (6) Cascaded Channel.

(1) Symmetric Channel

A Channel is said to be symmetric if the second & subsequent rows of the Channel matrix contain the same elements as that of first row but in different order.

eg: $P(Y/X) = \begin{bmatrix} \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ \frac{1}{3} & \frac{1}{6} & \frac{1}{2} \\ \frac{1}{6} & \frac{1}{2} & \frac{1}{3} \end{bmatrix}$

Channel Capacity of Symmetric Channel

W.K.T from the Channel matrix property,
Sum of all the elements of any row equal to unity
i.e., $\sum_{j=1}^q P(b_j/a_i) = 1$

In symmetric channel, "q" number of rows have the same elements but in different order

$$\therefore \sum_{j=1}^q p_j = 1$$

W.K.T Channel Capacity $C = \max\{I(A,B)\} \cdot q$

$$I(A,B) = H(B) - H(B/A)$$

The equivocation $H(B/A) = \sum_{i=1}^q \sum_{j=1}^q p(a_i, b_j) \log \frac{1}{p(b_j/a_i)}$

W.K.T Joint probability defⁿ $p(a_i, b_j) = p(a_i) \cdot p(b_j/a_i)$

$$H(B/A) = \sum_{i=1}^q \sum_{j=1}^s p(a_i) p(b_j/a_i) \log \frac{1}{p(b_j/a_i)}$$

$$= \sum_{i=1}^q \sum_{j=1}^s p(a_i) \cdot p_j \log \frac{1}{p_j}$$

$$= \left[\sum_{i=1}^q p(a_i) \right] \left[\sum_{j=1}^s p_j \log \frac{1}{p_j} \right]$$

w.k.t, $\sum_{i=1}^q p(a_i) = 1$ & $\sum_{j=1}^s p_j \log \frac{1}{p_j} = h.$

Where h is constant, which remains the same for all rows, since all rows consists of same elements.

$$\therefore H(B/A) = 1 \cdot h = h = \sum_{j=1}^s p_j \log \frac{1}{p_j}$$

$$\therefore I(A, B) = H(B) - H(B/A)$$

$$I(A, B) = H(B) - h$$

The channel capacity $C = \max [I(A, B)] \cdot \eta_k$; $\eta_k = 1 \text{ Sym/sec}$

$$\therefore C = \max [H(B) - h]$$

$$= \max H(B) - h.$$

$$\boxed{C = \log_2 s - h} \quad \text{Since } h \text{ is constant}$$

** P1. For the channel matrix given, compute the Channel Capacity with $\eta_s = 1000$ symbols/sec.

$$P(Y/X) = \begin{bmatrix} 0.6 & 0.2 & 0.2 \\ 0.2 & 0.6 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

Solⁿ:

$$C = (\log_2 8 - h) \eta_s$$

$$h = \sum_{i=1}^3 p_i \log \frac{1}{p_i}$$

$$h = 0.6 \log \frac{1}{0.6} + (0.2 \log \frac{1}{0.2}) 2$$

$$h = \underline{1.371} \text{ bits/sym}$$

$8 \rightarrow$ no. of output symbols = 3.

$$\therefore C = (\log_2 3 - 1.371) \cdot 1000$$

$$C = \underline{214} \text{ bits/sec}$$

** P2. A Channel has the following characteristics

$$P(Y/X) = \begin{bmatrix} 1/3 & 1/3 & 1/6 & 1/6 \\ 1/6 & 1/6 & 1/3 & 1/3 \end{bmatrix}$$

Find $H(X)$, $H(Y)$, $H(X, Y)$ & Channel capacity if $\eta = 1000$ symbols/sec.

Solⁿ: Since the i/p probabilities are not given, we will assume i/p's with the equal probabilities $p(x_1) = p(x_2) = 1/2$

$$\therefore p(x, y) = \begin{bmatrix} \frac{1}{6} & \frac{1}{6} & \frac{1}{12} & \frac{1}{12} \\ \frac{1}{12} & \frac{1}{12} & \frac{1}{6} & \frac{1}{6} \end{bmatrix}$$

from JPM property:

$$p(y_1) = p(y_2) = p(y_3) = p(y_4) = \frac{1}{6} + \frac{1}{12} = \frac{1}{4}$$

$$H(x) = \sum_{i=1}^L p(x_i) \log \frac{1}{p(x_i)}$$

$$H(x) = \left(\frac{1}{2} \log 2 \right) 2 = \underline{\underline{1 \text{ bits/sym}}}$$

$$H(y) = \sum_{i=1}^4 p(y_i) \log \frac{1}{p(y_i)}$$

$$H(y) = \left(\frac{1}{4} \log 4 \right) 4 = \underline{\underline{2 \text{ bits/sym}}}$$

$$\begin{aligned} H(x, y) &= \sum_{i=1}^2 \sum_{j=1}^4 p(x_i, y_j) \log \frac{1}{p(x_i, y_j)} \\ &= 4 \left(\frac{1}{6} \log 6 \right) + 4 \left(\frac{1}{12} \log 12 \right) \end{aligned}$$

$$H(x, y) = \underline{\underline{2.9183 \text{ bits/sym}}}$$

$$\text{Channel capacity: } C = (\log_2 8 - h) \text{ bits/s}$$

$$h = \sum_{j=1}^4 p_j \log \frac{1}{p_j} = 2 \left(\frac{1}{3} \log 3 \right) + 2 \left(\frac{1}{6} \log 6 \right)$$

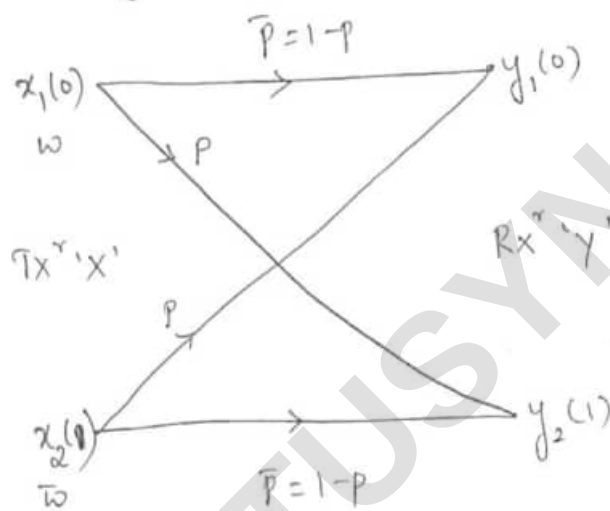
$$h = \underline{\underline{1.9183 \text{ bits/sym}}}$$

$$C = (\log_2 4 - 1.9183) 1000.$$

$$C = \underline{\underline{81.7 \text{ bits/sec}}}$$

(2) Binary Symmetric Channel (BSC)

BSC is one of the most commonly and widely used channels, whose channel diagram is



$$P(x_1) = w$$

$$P(x_2) = 1 - w = \bar{w}$$

P = probability of error
probability of reception of '1' when '0' is transmitted
probability of reception of '0' when '1' is transmitted.

$$P(Y/X) = \begin{bmatrix} \bar{P} & P \\ P & \bar{P} \end{bmatrix}$$

$$P(x_i) = \begin{bmatrix} w \\ \bar{w} \end{bmatrix}$$

$$H(Y/X) = h = \sum_{j=1}^2 P_j \log \frac{1}{P_j}$$

from the Symmetric channel derivation;
Channel Capacity of Symmetric channel is
 $C = (\log_2 2 - h) \eta_s ; \eta_s = 1$

$$C = (\log_2 2 - h) \cdot 98.$$

$$C = 1 - h \text{ bits/sec.}$$

P3. A binary symmetric channel has the following
** noise matrix with source probabilities $P(X_1) = \frac{2}{3}$ &

$$P(X_2) = \frac{1}{3}; \quad P(Y/X) = \begin{bmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{bmatrix}$$

- Determine $H(X)$, $H(Y)$, $H(X,Y)$, $H(Y/X)$, $H(X/Y)$ & $I(X,Y)$
- Find the channel capacity 'C'
- Find the channel efficiency & redundancy.

Solⁿ:

$$\begin{aligned} H(X) &= \sum_{i=1}^2 P(X_i) \log \frac{1}{P(X_i)} \\ &= \frac{2}{3} \log \frac{3}{2} + \frac{1}{3} \log 3 \\ H(X) &= \underline{0.9183 \text{ bits/sym}} \end{aligned}$$

$$P(X,Y) = P(Y/X) \cdot P(X) = \begin{bmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{bmatrix} \begin{bmatrix} 2/3 \\ 1/3 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/6 \\ 1/2 & 1/4 \end{bmatrix}$$

$$P(Y_1) = \frac{1}{2} + \frac{1}{12} = \frac{7}{12}; \quad P(Y_2) = \frac{1}{6} + \frac{1}{4} = \frac{5}{12}$$

$$\begin{aligned} H(Y) &= \sum_{j=1}^2 P(Y_j) \log \frac{1}{P(Y_j)} \\ &= \frac{7}{12} \log \frac{12}{7} + \frac{5}{12} \log \frac{12}{5} \\ H(Y) &= \underline{0.9799 \text{ bits/sym}} \end{aligned}$$

$$H(X, Y) = \sum_{i=1}^2 \sum_{j=1}^2 p(x_i, y_j) \log \frac{1}{p(x_i, y_j)}$$

$$= \frac{1}{2} \log_2 2 + \frac{1}{6} \log_2 6 + \frac{1}{12} \log_2 12 + \frac{1}{4} \log_2 4$$

$$H(X, Y) = \underline{\underline{1.7296 \text{ bits/sym}}}$$

$$H(Y/X) = h = p \log \frac{1}{p} + \bar{p} \log \frac{1}{\bar{p}}$$

$$= \frac{3}{4} \log \frac{4}{3} + \frac{1}{4} \log 4$$

$$H(Y/X) = h = \underline{\underline{0.8113 \text{ bits/sym}}}$$

$$H(X/Y) = H(X, Y) - H(Y)$$

$$= 1.7296 - 0.9799$$

$$H(X/Y) = \underline{\underline{0.7497 \text{ bits/sym}}}$$

$$I(X, Y) = H(X) - H(X/Y)$$

$$= 0.9183 - 0.7497$$

$$I(X, Y) = \underline{\underline{0.1686 \text{ bits/sym}}}$$

$$\text{Channel capacity; } C = 1 - h = 1 - H(Y/X)$$

$$C = 1 - 0.8113 = \underline{\underline{0.1887 \text{ bits/sec}}}$$

$$\text{Channel efficiency; } \eta_{ch} = \frac{I(X, Y)}{C} = \frac{0.1686}{0.1887} \times 100\%$$

$$\eta_{ch} = \underline{\underline{89.35\%}}$$

$$\text{Channel redundancy; } R_{ch} = \underline{\underline{10.65\%}}$$

* * P4. A binary channel has the following characteristics
 $P(Y/X) = \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{bmatrix}$. If the i/p symbols are transmitted with probabilities $3/4$ & $1/4$ respectively, find entropies $H(X)$, $H(X,Y)$ & $H(Y/X)$.

Solⁿ:

$$H(X) = \sum_{i=1}^2 p(x_i) \log \frac{1}{p(x_i)}$$

$$= \frac{3}{4} \log \frac{4}{3} + \frac{1}{4} \log 4$$

$$H(X) = \underline{0.8113 \text{ bits/sym}}$$

$$H(Y/X) = h = p \log \frac{1}{p} + \bar{p} \log \frac{1}{\bar{p}}$$

$$= \frac{2}{3} \log \frac{3}{2} + \frac{1}{3} \log 3$$

$$H(Y/X) = \underline{0.9183 \text{ bits/sym}}$$

$$H(X,Y) = H(X) + H(Y/X)$$

$$= 0.8113 + 0.9183$$

$$H(X,Y) = \underline{1.7296 \text{ bits/sym}}$$

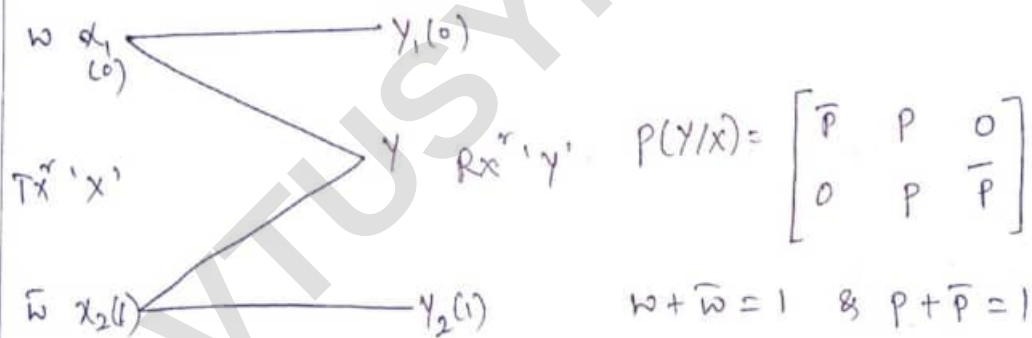
3. Binary Erasure Channel (BEC)

BEC is one of the important channel used in digital communication. Whenever an error occurs, the symbol will be received as 'y' & no decision will be made about the information, but an immediate request will be made through reverse channel.

The transmission of the signal takes place till a correct symbol is received at the output. This ensures 100% correct data recovery. Since the error is totally erased in this type of channel, it is called "binary erasure channel".

The disadvantage with this is the requirement of a reverse channel.

BEC is also a symmetric channel whose Channel matrix can be constructed from the Channel diagram. Channel diagram is shown below.



$$H(Y/X) = h = \sum_{j=1}^2 p_j \log \frac{1}{p_j} = P \log \frac{1}{P} + \bar{P} \log \frac{1}{\bar{P}}$$

$$C = \max [I(X, Y)] \text{ bits/s} = \max [H(X) - H(X/Y)] \text{ bits/s}$$

$$H(X) = \sum_{i=1}^2 P(x_i) \log \frac{1}{P(x_i)}$$

$$H(X) = w \log \frac{1}{w} + \bar{w} \log \frac{1}{\bar{w}} \quad P(X) = \begin{bmatrix} w \\ \bar{w} \end{bmatrix}$$

$$P(x, y) = \begin{bmatrix} \bar{p}w & pw & 0 \\ 0 & p\bar{w} & \bar{p}\bar{w} \end{bmatrix} \quad \because P(x, y) = P(y/x) \cdot P(x)$$

property of JPM

$$P(y_1) = \bar{p}w$$

$$P(y_2) = pw + p\bar{w} = p(w + \bar{w})$$

$$P(y) = p$$

$$P(y_2) = \bar{p}\bar{w}$$

$$P(x/y) = \frac{P(x, y)}{P(y)}$$

$$P(x/y) = \begin{bmatrix} \bar{p}w/\bar{p}w & \frac{pw}{p} & 0 \\ 0 & \frac{p\bar{w}}{p} & \frac{\bar{p}\bar{w}}{\bar{p}\bar{w}} \end{bmatrix}$$

$$P(x/y) = \begin{bmatrix} 1 & w & 0 \\ 0 & \bar{w} & 1 \end{bmatrix}$$

$$H(x/y) = \sum_{i=1}^2 \sum_{j=1}^3 P(x_i, y_j) \log \frac{1}{P(x_i/y_j)}$$

$$= \bar{p}w \log(1) + pw \log \frac{1}{w} + 0 + 0 + p\bar{w} \log \frac{1}{\bar{w}} + \bar{p}\bar{w} \log(1)$$

$$= p(w \log \frac{1}{w} + \bar{w} \log \frac{1}{\bar{w}})$$

$$\text{Mutual information; } I(x, y) = H(x) - H(x/y)$$

$$I(x, y) = (w \log \frac{1}{w} + \bar{w} \log \frac{1}{\bar{w}}) - p(w \log \frac{1}{w} + \bar{w} \log \frac{1}{\bar{w}})$$

$$I(x, y) = (1-p)(w \log \frac{1}{w} + \bar{w} \log \frac{1}{\bar{w}})$$

$$I(x, y) = \bar{p} \cdot H(x)$$

$$\text{Channel Capacity } C = \max [I(X, Y)] \cdot g_s ; g_s = 1$$

$$C = \max [\bar{p} \cdot H(X)]$$

$$C = \bar{p} \cdot H(X)_{\max}$$

$$C = \bar{p} \cdot \log_2^2 ; H(X)_{\max} = \log_2^2 = 1$$

$$\boxed{C = \bar{p}}$$

Even though BEC is symmetric channel, we cannot consider $I(X, Y) = H(Y) - H(Y/X)$, since $H(Y)$ involves 'Y' which is rejected at the receiver.

A) Noisless Channel

A channel represented by a channel matrix with one & only one nonzero element in every column is defined as noisless channel.

The noisless channel matrix has two properties.

- (1) Each column contains only one element with other elements in that column being zeros.
- (2) Sum of all the elements in any row is equal to unity.

eg:

$$P(B/A) = \begin{bmatrix} 1/2 & 1/2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3/5 & 3/10 & 1/10 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Channel capacity of a noiseless channel.

$$C = \max [I(A, B)] \cdot \eta_s$$

$$= \max [H(A) - H(A|B)]$$

$$= \max H(A) - 0 \quad \because H(A|B) = 0, \text{ since we have zeros in the matrix}$$

$$C = H(A)_{\max}$$

$$C = \log_2 \eta_s \text{ bits/sec.}$$

(5) Deterministic channel

A channel represented by a channel matrix with one & only one non-zero element in every row is defined as deterministic channel.

A deterministic channel must satisfy the following two properties.

- (1) The sum of the elements in every row must be equal to unity.
- (2) Each row must contain only one entry & other elements in that row must be zero.

eg: $P(B/A) = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Channel capacity of a deterministic channel.

$$C = \max [I(A, B)]$$

$$= \max [H(B) - H(B|A)]$$

$H(B/A) = 0$, since we have 0 & 1 elements in matrix

$$C = \max \{ HCB - 0 \} \cdot 918 ; 918 = 1$$

$$C = HCB_{\max}$$

$$C = \log_2 8 \text{ bits/sec}$$

(6) Cascaded Channel

Let us consider two channels connected in cascade as shown in figure. When the information is transmitted from X to Y through Channel I, there will be loss of information due to noise in Channel I & the mutual information at the output of the Channel I is

$$I(X, Y) = H(Y) - H(Y/X)$$



When the information is passed through Channel II there will be further loss of information & the mutual information at the output of Channel II would be

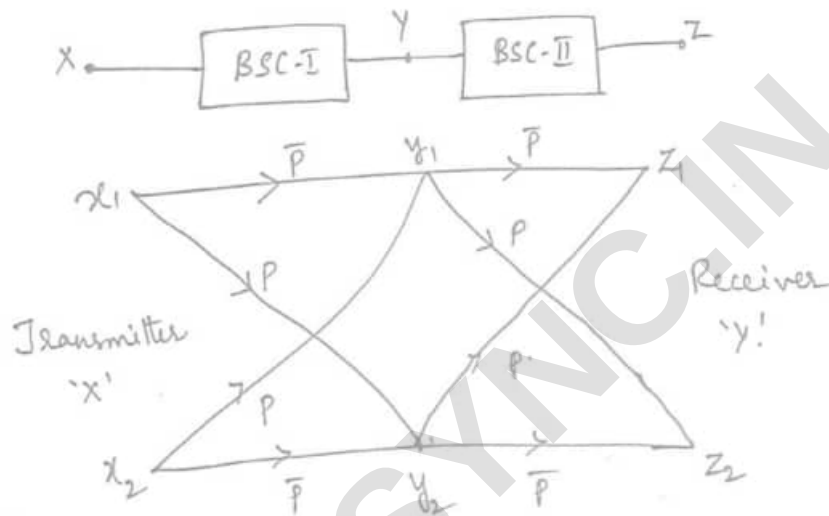
$$I(X, Z) = H(Z) - H(Z/X)$$

Therefore we can conclude that

$$I(X, Z) < I(X, Y)$$

$$\text{also } C(X, Z) < C(X, Y)$$

P5. Two binary symmetric channels are connected in cascade as shown in figure. Find the overall channel capacity of cascaded connection assuming that both the channels have same channel diagram with $P=0.2$. Also show that $C_{XZ} < C_{XY}$



Sol:

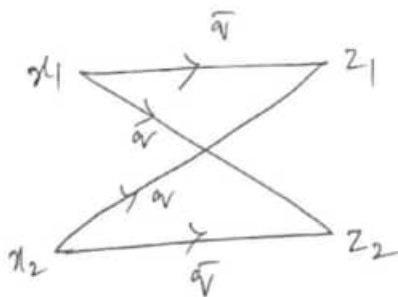
$$P(Z_1/x_1) = \bar{P} + P^2 = (\bar{P} + P)^2 - 2P\bar{P} \quad \text{Assume } 2P\bar{P} = q$$

$$P(Z_1/x_1) = 1 - q = P(Z_2/x_2) = \bar{q}$$

$$P(Z_1/x_2) = P\bar{P} + P\bar{P} = 2P\bar{P} = q$$

$$P(Z_1/x_2) = P(Z_2/x_1) = q$$

∴ Overall Channel diagram & matrix is shown below



$$P(z/x) = \begin{bmatrix} \bar{q} & q \\ q & \bar{q} \end{bmatrix}$$

The overall channel capacity of BSC with $\eta_s = 1$

$$C_{xz} = (1-h) \eta_s = 1 - \left[q \log \frac{1}{q} + \bar{q} \log \frac{1}{\bar{q}} \right]$$

$$q = 2p\bar{p}$$

$$p = 0.2 \text{ \& } \bar{p} = 0.8$$

$$q = 2 \times 0.2 \times 0.8 = 0.32$$

$$q = \underline{\underline{0.32}}$$

$$\bar{q} = 1 - q = 1 - 0.32 = \underline{\underline{0.68}}$$

$$\therefore C_{xz} = 1 - \left[0.68 \log \frac{1}{0.68} + 0.32 \log \frac{1}{0.32} \right]$$

$$C_{xz} = \underline{\underline{0.0956}} \text{ bits/sec}$$

$$C_{xy} = (1-h) \eta_s ; \quad \eta_s = 1$$

$$= 1 - \left(p \log \frac{1}{p} + \bar{p} \log \frac{1}{\bar{p}} \right)$$

$$= 1 - \left(0.2 \log \frac{1}{0.2} + 0.8 \log \frac{1}{0.8} \right)$$

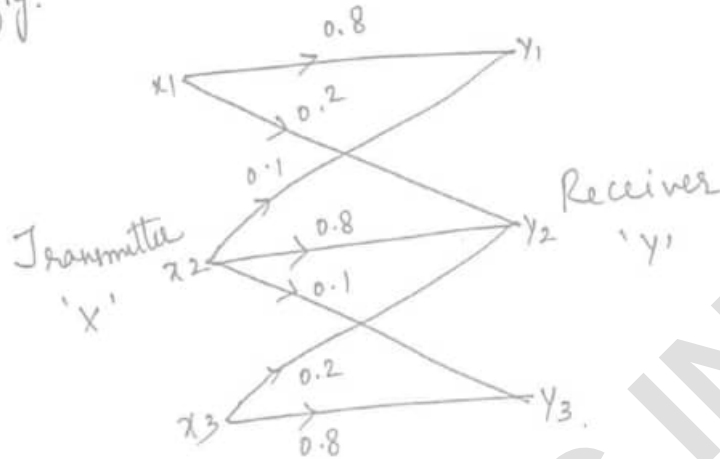
$$C_{xy} = \underline{\underline{0.278}} \text{ bits/sec}$$

$$\therefore C_{xz} < C_{xy}$$

$$\underline{\underline{0.0956}} < 0.278 \text{ verified.}$$

Estimation of Channel Capacity by Muroga's method

P6. Find the capacity of the discrete channel shown in fig.



Sol: From the channel diagram, we can write the channel

matrix as
$$p(Y/X) = \begin{bmatrix} 0.8 & 0.2 & 0 \\ 0.1 & 0.8 & 0.1 \\ 0 & 0.2 & 0.8 \end{bmatrix}$$

Above channel matrix does not belong to any of the special channel, therefore channel capacity has to be calculated using Muroga's method.

$$\begin{bmatrix} 0.8 & 0.2 & 0 \\ 0.1 & 0.8 & 0.1 \\ 0 & 0.2 & 0.8 \end{bmatrix} \begin{bmatrix} Q_1 \\ Q_2 \\ Q_3 \end{bmatrix} = \begin{bmatrix} 0.8 \log 0.8 + 0.2 \log 0.2 \\ 0.8 \log 0.8 + 2(0.1 \log 0.1) \\ 0.8 \log 0.8 + 0.2 \log 0.8 \end{bmatrix} = \begin{bmatrix} -0.722 \\ -0.922 \\ -0.722 \end{bmatrix}$$

$$0.8 Q_1 + 0.2 Q_2 = -0.722$$

$$0.1 Q_1 + 0.8 Q_2 + 0.1 Q_3 = -0.922$$

$$0.2 Q_2 + 0.8 Q_3 = -0.722$$

Solving the eq^{ns} $Q_1 = -0.6553$, $Q_2 = -0.9887$ & $Q_3 = -0.6553$.

Channel Capacity ; $C = \log_2 [2^{0.1} + 2^{0.2} + 2^{0.3}]$ nB

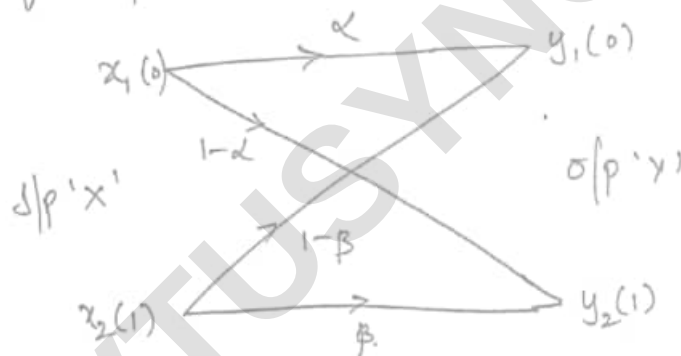
$$C = \log_2 [2^{-0.6553} + 2^{-0.9887} + 2^{-0.6553}] ; n_B = 1$$

$$C = 0.8269 \text{ bits/sec}$$

P7. A non-symmetric binary channel shown in fig.

(i) Find $H(X)$, $H(Y)$, $H(X/Y)$ & $H(Y/X)$ given $P(X=0) = \frac{1}{4}$
 $P(X=1) = \frac{3}{4}$, $\alpha = 0.75$, $\beta = 0.9$

(ii) Find the capacity of the binary symmetric channel
if $\alpha = \beta = 0.75$



Soln: $P(Y/X) = \begin{bmatrix} \alpha & 1-\alpha \\ 1-\beta & \beta \end{bmatrix} = \begin{bmatrix} 0.75 & 0.25 \\ 0.1 & 0.9 \end{bmatrix}$

$$P(X_1) = \frac{1}{4} ; P(X_2) = \frac{3}{4}$$

$$P(X, Y) = P(Y/X) \cdot P(X) = \begin{bmatrix} 0.75 & 0.25 \\ 0.1 & 0.9 \end{bmatrix} \begin{bmatrix} 1/4 \\ 3/4 \end{bmatrix}$$

$$P(X, Y) = \begin{bmatrix} 0.1875 & 0.0625 \\ 0.075 & 0.675 \end{bmatrix}$$

$$H(X) = \sum_{i=1}^2 p(x_i) \log \frac{1}{p(x_i)}$$

$$= \frac{1}{4} \log 4 + \frac{3}{4} \log \frac{4}{3}$$

$$H(X) = \underline{\underline{0.8113 \text{ bits/sym}}}$$

$$P(Y_1) = 0.2625 = 0.1875 + 0.075$$

$$P(Y_2) = 0.0625 + 0.675 = 0.7375$$

$$H(Y) = \sum_{j=1}^2 p(y_j) \log \frac{1}{p(y_j)}$$

$$= 0.2625 \log \frac{1}{0.2625} + 0.7375 \log \frac{1}{0.7375}$$

$$H(Y) = \underline{\underline{0.8305 \text{ bits/sym}}}$$

$$H(Y/X) = \sum_{i=1}^2 \sum_{j=1}^2 p(x_i, y_j) \log \frac{1}{p(y_j/x_i)}$$

$$= 0.1875 \log \frac{1}{0.75} + 0.0625 \log \frac{1}{0.25} + 0.075 \log \frac{1}{0.1}$$

$$+ 0.675 \log \frac{1}{0.9}$$

$$H(Y/X) = \underline{\underline{0.5546 \text{ bits/sym}}}$$

$$H(X, Y) = H(Y, X)$$

$$H(Y) + H(X/Y) = H(X) + H(Y/X)$$

$$H(X/Y) = H(X) + H(Y/X) - H(Y)$$

$$= 0.8113 + 0.5546 - 0.8305$$

$$H(X/Y) = \underline{\underline{0.5354 \text{ bits/sym}}}$$

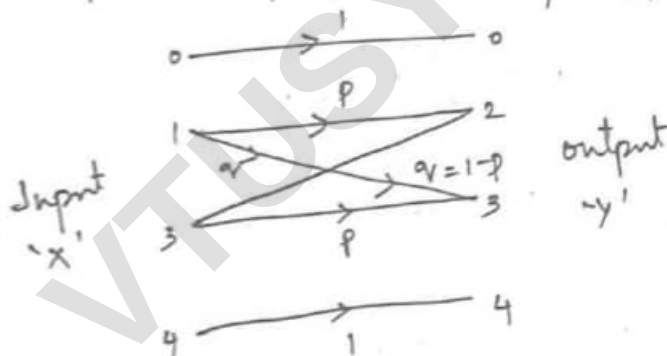
(ii) When $\alpha = \beta = 0.75$, then the Channel Capacity of BSC, $C = 1 - [p \log \frac{1}{p} + \bar{p} \log \frac{1}{\bar{p}}] \eta_b$.

$$C = 1 - \left[0.75 \log \frac{1}{0.75} + 0.25 \log \frac{1}{0.25} \right]$$

$$C = 0.1887 \text{ bits/sec} \quad \eta_b = 1$$

P8. Show that the capacity of a discrete memoryless Channel shown in fig. is $C = \log_2 \left[\frac{2(1+\beta)}{\beta} \right]$ bps where $\beta = 2^\alpha$ & $\alpha = -[p \log_2 p + q \log_2 q]$, symbol rate $\eta_b = 1$ bits/sym.

$P(X=0) = P(X=3) = p$, $P(X=1) = P(X=2) = q$.



Sol: Channel matrix is given by

$$P(Y/X) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & p & q & 0 \\ 0 & q & p & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

By Muroga's method;

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & p & q & 0 \\ 0 & q & p & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \begin{bmatrix} 1 \log 1 \\ p \log p + q \log q \\ q \log q + p \log p \\ 1 \log 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -\alpha \\ -\alpha \\ 0 \end{bmatrix}$$

$$\text{since } -[p \log p + q \log q] = +\alpha$$

$$Q_1 = 0 \quad \& \quad Q_4 = 0$$

$$pQ_2 + qQ_3 = -\alpha$$

$$qQ_2 + pQ_3 = -\alpha$$

$$\text{we can write } pQ_2 + qQ_3 = qQ_2 + pQ_3$$

$$Q_2(p - q) = Q_3(p - q)$$

$$\underline{\underline{Q_2 = Q_3}}$$

$$pQ_2 + qQ_3 = -\alpha$$

$$pQ_2 + qQ_2 = -\alpha$$

$$Q_2(p + q) = -\alpha$$

$$Q_2 = -\alpha$$

$$q = 1 - p \quad \& \quad p + q = 1$$

$$\therefore \underline{\underline{Q_2 = Q_3 = -\alpha}}$$

$$\text{Channel capacity ; } C = \log [2^{Q_1} + 2^{Q_2} + 2^{Q_3} + 2^{Q_4}] \cdot 913$$

$$C = \log_2 [2^0 + 2^{-\alpha} + 2^{-\alpha} + 2^0] \quad 913$$

$$C = \log_2 \left[2 + 2 \cdot \frac{1}{\beta} \right] \quad \because \beta = 2^\alpha$$

$$C = \log_2 \left[2 \left(1 + \frac{1}{\beta} \right) \right]$$

$$\boxed{C = \log_2 \left[\frac{2(1 + \beta)}{\beta} \right]} \quad \text{proved}$$

Continuous Channel.

In this continuous channel, the input signals are continuous functions of time & the function of the Channel is to produce at its output the electrical waveform presented at its input.

Channel Capacity Theorem (Shannon-Hartley Law) Shannon Third Theorem.

The waveform received at the receiver may be accompanied by a waveform which varies with respect to time in an entirely unpredictable manner. This unpredictable voltage waveform is a random process called noise.

In channels, the noise is almost white and it has a distribution which resembles Gaussian (or normal) distribution with zero mean & some variance. Hence, this noise is also called "additive white Gaussian noise (AWGN)".

Shannon Hartley law also called Shannon's Third theorem states that the capacity of a band limited Gaussian channel with AWGN is given by

$$C = B \log_2 \left(1 + \frac{S}{N} \right) \text{ bits/sec} \text{ where } B \rightarrow \text{Channel Bandwidth in Hz}$$

$S \rightarrow$ Signal power in watts

$N \rightarrow$ Noise power in watts $= \eta B$

where the 2 sided power spectral density of noise is $(\eta/2)$ watts/Hz

* P1. A CRT terminal is used to enter alphanumeric data into a computer. The CRT is connected through voice-grade telephone line having usable bandwidth of 3KHz & an output (S/N) of 10 dB. Assume that the terminal has 128 characters & data is sent in an independent manner with equal probability

- (i) Find the average information per character
- (ii) Find the capacity of the channel.
- (iii) Find the maximum rate at which data can be sent from terminal to the computer without error.

Solⁿ: (i) Since all the characters are equiprobable, the average information content is

$$H = H_{\max} = \log_2 q = \log_2 128$$

$$H_{\max} = 7 \text{ bits/character.}$$

$$(ii) \quad C = B \log_2 \left(1 + \frac{S}{N} \right) \quad 10 \log_{10} \frac{S}{N} = 10$$

$$C = 3 \times 10^3 \log_2 (1 + 10) \quad \log_{10} \left(\frac{S}{N} \right) = 1$$

$$C = \underline{10378.295 \text{ bits/sec}} \quad \frac{S}{N} = 10^1 = 10$$

(iii) Avg. information rate; $R_s = q_s \cdot H$
for error free transmission, $R_s \leq C$
 $q_s \cdot H \leq C$

$$q_s < \frac{C}{H} = \frac{10378.295}{7}$$

$$q_s < 1482.614 \text{ character/sec.}$$

$\therefore$ The max. rate at which data can be sent from terminal to computer without error is 1482.614 character/sec.

* P2. An analog signal has a 4KHz bandwidth. The signal is sampled at 2.5 times the Nyquist rate & each sample quantized into 256 equally likely levels. Assume that the successive samples are statistically independent.

(i) Find the information rate of this source.
(ii) Can the output of this source be transmitted without errors over a Gaussian channel of bandwidth 50 KHz & (S/N) ratio of 20dB?

(iii) If the output of this source is to be transmitted without errors over an analog channel having (S/N) of 10 dB, compute the bandwidth requirement of the channel.

Solⁿ (i) $H = H_{\max} = \log_2 Q = \log_2 256 = 8 \text{ bits/levels}$
 $R_s = f_s \cdot H$ sampling rate $f_s = (2.5) \text{ Nyquist rate}$

$$R_s = 20,000 \times 8$$

$$R_s = 160 \times 10^3 \text{ bits/sec}$$

$$f_s = 2.5 \times (2B)$$

$$f_s = 2.5 \times 2 \times 4K$$

$$f_s = 20000 \text{ samples/sec.}$$

(ii) $C = B \log_2 \left(1 + \frac{S}{N}\right)$ $10 \log_{10} \frac{S}{N} = 20$

$$C = 50 \times 10^3 \log_2 (1 + 100) \quad \log_{10} \frac{S}{N} = 2 \quad ; \quad \frac{S}{N} = 10^2 = 100$$

$$C = 332.91 \times 10^3 \text{ bits/sec}$$

Since $R_s < C$, according to Shannon's second theorem, it is possible to transmit over the channel without error.

(iii) $C = B \log \left(1 + \frac{S}{N}\right) = R_s$

$$B = \frac{R_s}{\log \left(1 + \frac{S}{N}\right)} = \frac{160 \times 10^3}{\log (1 + 10)}$$

$$B = 46.25 \text{ KHz}$$

$$10 \log_{10} \frac{S}{N} = 10$$

$$\log_{10} \frac{S}{N} = 1$$

$$\frac{S}{N} = 10.$$